



Oxford Cambridge and RSA

# Wednesday 17 May 2023 – Morning

## AS Level Physics A

### H156/01 Breadth in physics

**Time allowed: 1 hour 30 minutes**



**You must have:**

- the Data, Formulae and Relationships Booklet

**You can use:**

- a scientific or graphical calculator
- a ruler (cm/mm)



Please write clearly in black ink. **Do not write in the barcodes.**

Centre number

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Candidate number

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First name(s)

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Last name

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### INSTRUCTIONS

- Use black ink. You can use an HB pencil, but only for graphs and diagrams.
- Write your answer to each question in the space provided. If you need extra space use the lined pages at the end of this booklet. The question numbers must be clearly shown.
- Answer **all** the questions.
- Where appropriate, your answer should be supported with working. Marks might be given for using a correct method, even if your answer is wrong.

### INFORMATION

- The total mark for this paper is **70**.
- The marks for each question are shown in brackets [ ].
- This document has **24** pages.

### ADVICE

- Read each question carefully before you start your answer.

## 2

## Section A

You should spend a maximum of **25 minutes** on this section.

Write your answer to each question in the box provided.

- 1 What is a typical value, in cm, for the wavelength of microwave radiation?

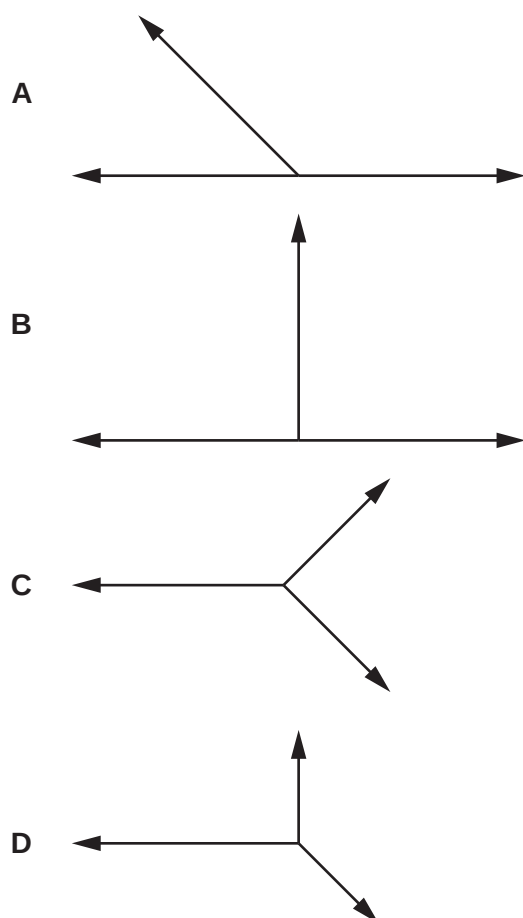
- A 0.0003
- B 0.03
- C 3
- D 300

Your answer

[1]

- 2 The diagrams show four systems of forces with three forces acting at a single point. The forces are in the same plane. The diagrams are drawn to scale.

Which system could be in equilibrium?



Your answer

[1]

3

- 3 An electron beam is passed through a thin slice of graphite and a diffraction pattern is produced.

The approximate spacing between the carbon atoms in the graphite is  $d$ .

The approximate de Broglie wavelength of an electron in the beam is  $\lambda$ .

Which statement is correct?

A  $\lambda \approx d$

B  $\lambda \approx \frac{1}{d}$

C  $\lambda \gg d$

D  $\lambda \ll d$

Your answer

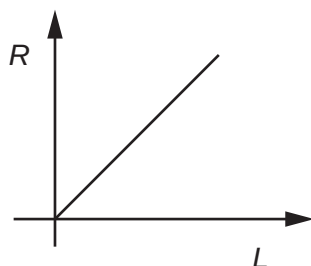
☐

[1]

- 4 A student investigates a conducting wire of constant cross-sectional area, at constant temperature.

The resistance  $R$  is measured for a range of lengths  $L$ .

The following graph is plotted:



Which expression is equal to the gradient of the graph?

A resistivity

B resistivity  $\times$  cross-sectional area

C  $\frac{\text{resistivity}}{\text{cross-sectional area}}$

D  $\frac{\text{cross-sectional area}}{\text{resistivity}}$

Your answer

☐

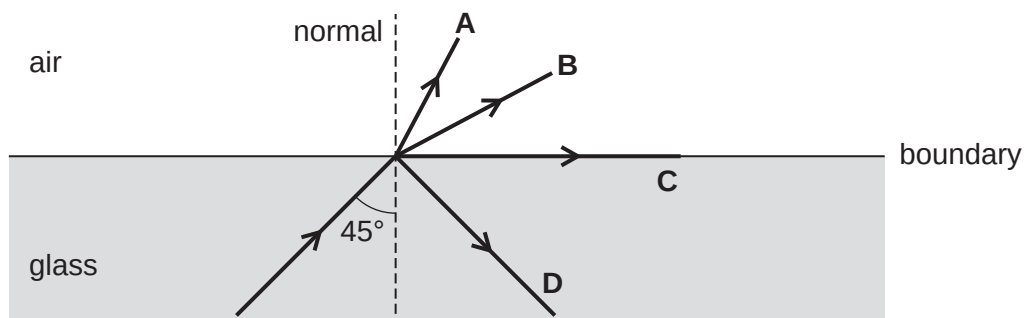
[1]

4

- 5 A ray of monochromatic light is travelling through glass.

The refractive index of the glass is 1.5.

The ray is incident on a glass-air boundary with angle of incidence of  $45^\circ$ .



Which is the correct path of the ray after it reaches the boundary?

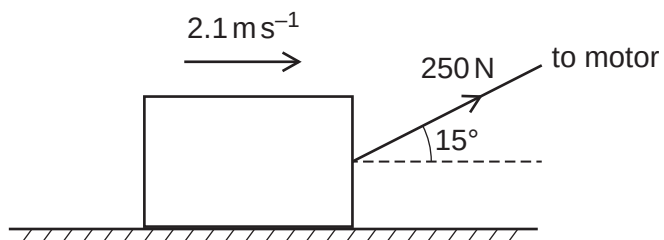
Your answer ☐

[1]

- 6 The diagram shows a motor pulling a load along a flat, horizontal surface.

The load is connected to the motor by a string at an angle of  $15^\circ$  to the surface. The tension in the string is 250 N. The load reaches a constant speed of  $2.1 \text{ m s}^{-1}$ .

The diagram does not show the other forces acting on the load.



What is the output power of the motor?

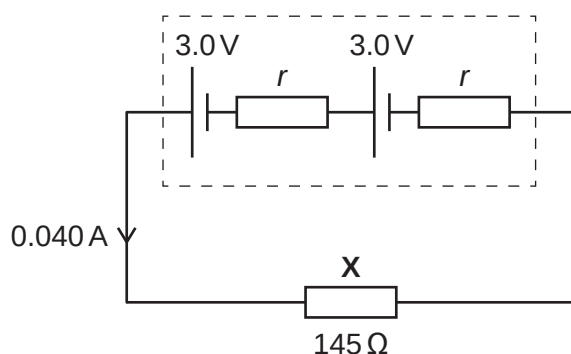
- A 120 W
- B 140 W
- C 460 W
- D 510 W

Your answer ☐

[1]

5

- 7 Two identical cells are connected in series in a circuit with a resistor **X** of resistance  $145\ \Omega$ . The e.m.f. of each cell is  $3.0\text{ V}$ . The current in **X** is  $0.040\text{ A}$ .



What is the internal resistance  $r$  of one of the cells?

- A  $0.10\ \Omega$
- B  $0.20\ \Omega$
- C  $2.5\ \Omega$
- D  $5.0\ \Omega$

Your answer

[1]

- 8 Electromagnetic radiation is incident on a clean metal plate. Electrons are released from the surface of the plate.

Which statement is correct?

- A The energy of a photon is directly proportional to its frequency.
- B The kinetic energy of the released electrons depends on the rate of incidence of photons.
- C The photoelectric effect demonstrates the wave-like nature of electromagnetic radiation.
- D The rate at which electrons are released depends on the frequency of the radiation.

Your answer

[1]

## 6

- 9 Two copper wires, **X** and **Y**, are connected in series to a source of e.m.f.

The length of **X** is equal to the length of **Y**.

The cross-sectional area of **X** is greater than the cross-sectional area of **Y**.

Which **two** quantities are equal in value for both **X** and **Y**?

- A charge carrier density, current
- B charge carrier density, electron drift velocity
- C current, resistance
- D electron drift velocity, resistance

Your answer

☐

[1]

- 10 Resistance and resistivity are two quantities used to describe the behaviour of a conductor.

How do these quantities change, if at all, when the length of a conductor is increased?

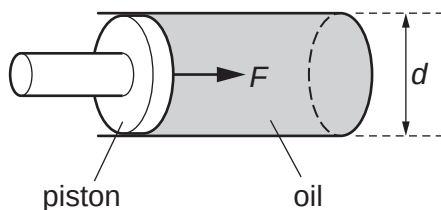
|   | resistance | resistivity |
|---|------------|-------------|
| A | constant   | constant    |
| B | constant   | increase    |
| C | increase   | constant    |
| D | increase   | increase    |

Your answer

☐

[1]

- 11 The diagram shows a closed cylinder with internal diameter  $d$ .



The cylinder is filled with oil. A piston applies a force  $F$  to the oil.

Which is the correct expression for the pressure in the oil?

- A  $\frac{2F}{\pi d^2}$
- B  $\frac{\pi d^2}{2F}$
- C  $\frac{4F}{\pi d^2}$
- D  $\frac{\pi d^2}{4F}$

Your answer

☐

[1]

- 12 An LED is designed to emit red light. In an experiment, the potential difference across the LED is gradually increased. It begins to emit light when the potential difference reaches a value  $V_0$ .

The experiment is repeated for an LED designed to emit green light. A different value of  $V_0$  is observed.

Which row of the table describes the values in the second experiment?

|   | change to $V_0$ | change to wavelength of emitted light |
|---|-----------------|---------------------------------------|
| A | decrease        | increase                              |
| B | decrease        | decrease                              |
| C | increase        | increase                              |
| D | increase        | decrease                              |

Your answer

☐

[1]

8

- 13** Unpolarised light is observed through a single polarising filter.

The intensity of the light transmitted by the filter is half the intensity of the incident light.

What happens to the intensity of the transmitted light when the filter is rotated through  $90^\circ$ ?

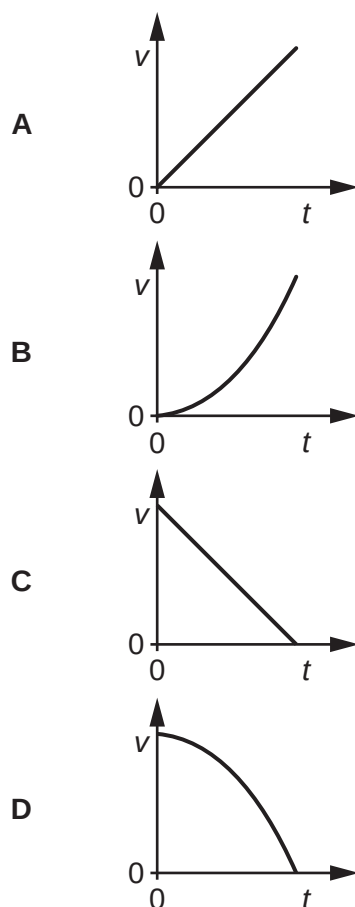
- A** decreases
- B** decreases and then increases
- C** increases
- D** does not change

Your answer

[1]

- 14** An object is released from rest and then falls freely under gravity. Air resistance is negligible.

Which graph shows how the velocity  $v$  of the object varies with time  $t$ ?



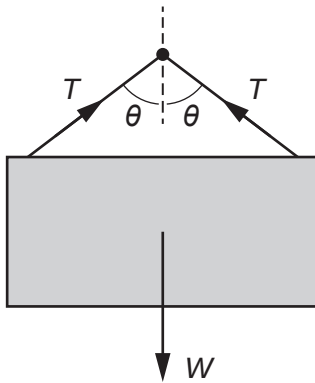
Your answer

[1]



9

- 15** A cable is used to hang a picture from a nail. The diagram shows all the forces acting on the picture.  $T$  is the tension in the cable and  $W$  is the weight of the picture.



Which is the correct expression for  $W$ ?

- A**  $W = T \cos \theta$
- B**  $W = 2T \cos \theta$
- C**  $W = T \sin \theta$
- D**  $W = 2T \sin \theta$

Your answer

[1]

- 16** An object is in equilibrium.

Only two forces, **X** and **Y**, act on the object.

Which of the following statements must be correct?

- 1 **X** and **Y** are equal and opposite.
- 2 **X** and **Y** are a Newton's 3rd law force pair.
- 3 The object is at rest.

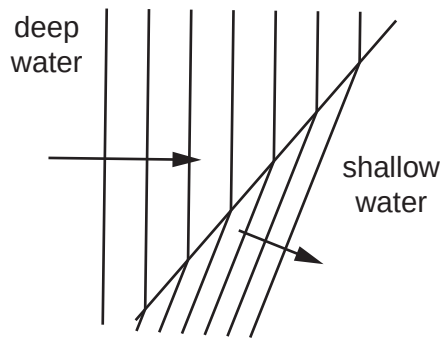
- A** Only 1
- B** 1 and 2
- C** 1 and 3
- D** 1, 2 and 3

Your answer

[1]

10

- 17 A ripple tank can be used to demonstrate wave effects. When water waves in a ripple tank travel from deep to shallow water the wavelength decreases.



What happens to the speed and frequency of the waves as they move from deep to shallow water?

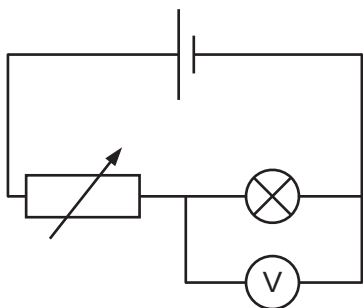
|          | speed    | frequency |
|----------|----------|-----------|
| <b>A</b> | decrease | increase  |
| <b>B</b> | decrease | constant  |
| <b>C</b> | increase | increase  |
| <b>D</b> | increase | constant  |

Your answer

[1]

11

- 18 A tungsten filament lamp is connected in a circuit as shown.



The variable resistor is adjusted so that the temperature of the filament lamp increases.

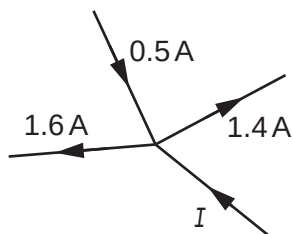
What happens to the resistance of the lamp and the reading on the voltmeter?

|          | resistance of the lamp | voltmeter reading |
|----------|------------------------|-------------------|
| <b>A</b> | decreases              | decreases         |
| <b>B</b> | decreases              | increases         |
| <b>C</b> | increases              | decreases         |
| <b>D</b> | increases              | increases         |

Your answer

[1]

- 19 The diagram shows the currents passing in and out of a point in a circuit.



What is the value of the current  $I$ ?

- A** 0.3A  
**B** 0.5A  
**C** 2.5A  
**D** 3.0A

Your answer

[1]

12

- 20 An uncharged oil drop gains a charge of  $+4.0 \times 10^{-18} \text{ C}$ .

What is the change in the number of electrons on the oil drop?

- A Gained 25 electrons
- B Gained 64 electrons
- C Lost 25 electrons
- D Lost 64 electrons

Your answer

[1]

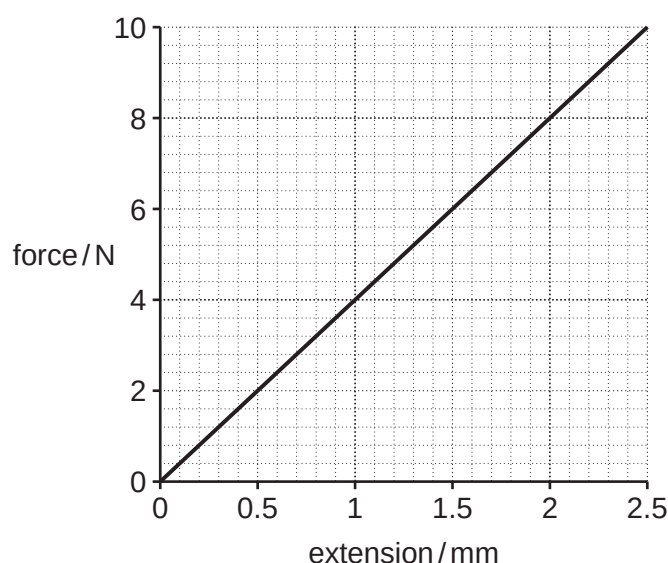
**13**  
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14

## Section B

- 21 A student investigates the properties of a spring. They hang masses from the spring and measure the extension of the spring for every 100 g mass added. They plot their results on a graph of force against extension.



- (a) Calculate the force constant of the spring.

force constant = .....  $\text{N m}^{-1}$  [2]

- (b) The student extends their investigation to determine the Young modulus of copper. They make measurements to determine the length and the cross-sectional area of a copper wire.

Describe how they should determine the cross-sectional area of the wire.

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.....  
.....  
.....  
.....  
..... [3]

15

- (c) (i) Show that an expression for the Young modulus  $E$  is

$$E = \frac{kL}{A}$$

$L$  = length of wire

$A$  = cross-sectional area of wire

$k$  = force constant of wire

[2]

- (ii) The student records the following results for a copper wire.

$$L = 2.0 \text{ m} \pm 0.05 \%$$

$$A = 2.9 \times 10^{-8} \text{ m}^2 \pm 2 \%$$

$$k = 1670 \text{ N m}^{-1} \pm 1.25 \%$$

Calculate the value of the Young modulus of the wire and its percentage uncertainty.

Young modulus = .....  $\text{N m}^{-2} \pm$  ..... % [3]

- (iii) The student researches the Young modulus of copper.

They find a value of  $1.17 \times 10^{11} \text{ N m}^{-2}$ .

Determine whether this value is consistent with your answer to (c)(ii).

[2]

**22** Sound waves in air are longitudinal waves consisting of compressions and rarefactions.

**(a)** Explain how the movement of air molecules creates compressions and rarefactions.

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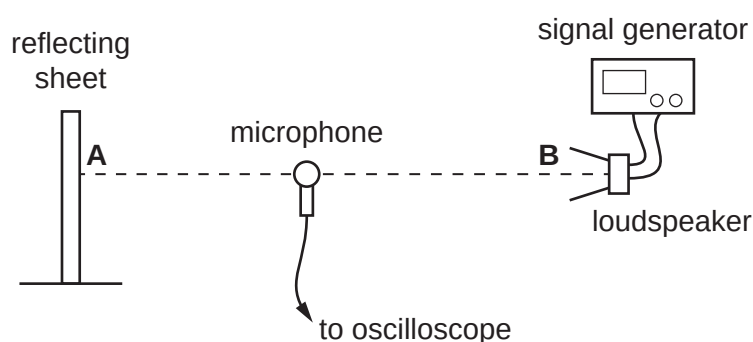
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..... **[3]**

**(b)** A student investigates sound waves. They set up the following apparatus.



The sound wave emitted from the loudspeaker at **B** travels to the reflecting sheet at **A** and is reflected. A stationary wave is formed between the loudspeaker and the sheet.

The student moves a microphone along the line **AB**. The microphone is connected to an oscilloscope. The oscilloscope shows the relative amplitude of the stationary wave at each point along the line. The student observes a series of nodes and antinodes.

**(i)** Explain how a stationary wave with nodes and antinodes is formed.

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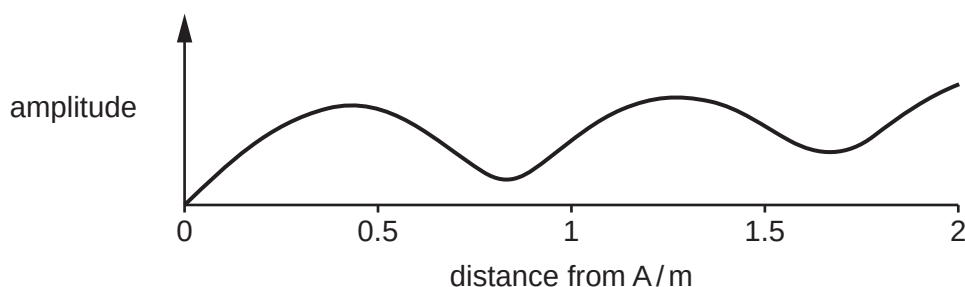
.....

..... **[3]**



17

- (ii) The student measures the amplitude of the stationary wave at a range of distances from the reflecting sheet **A**. Their results are shown below.



The amplitudes at the nodes are observed to be:

- not exactly equal to zero
- closer to zero at distances closer to the reflecting sheet.

Explain these observations.

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..... [3]

- (iii) The student measures the distance between two adjacent nodes as 0.84 m.

The frequency of the sound wave is 200 Hz.

Use these measurements to calculate a value for the speed of sound waves in air.

speed of sound waves = .....  $\text{m s}^{-1}$  [2]

- (iv) The student wants to reduce the uncertainty in their calculated value for the speed of sound waves in air.

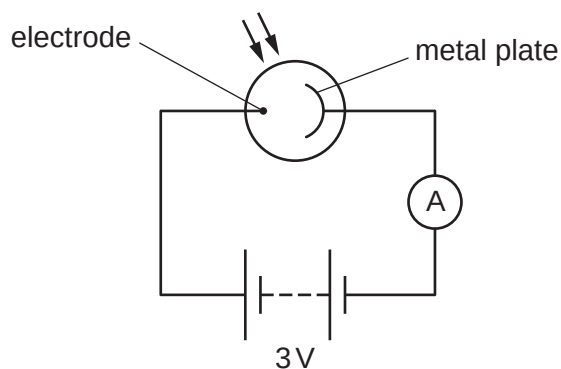
Suggest a suitable improvement to the student's method.

.....

..... [1]

18

- 23** A light meter is used to measure the intensity of electromagnetic radiation. The meter consists of a metal plate and an electrode within an evacuated glass tube. It is connected to a circuit with an ammeter, a battery of e.m.f. 3.0 V and negligible internal resistance.



Electromagnetic radiation is incident on the metal plate. Electrons are released due to the photoelectric effect and are attracted to the electrode.

- (a)** Calculate the work done on an electron as it moves from the metal plate to the electrode.

work done = ..... J [2]

- (b)** Explain why the frequency of the electromagnetic radiation must be above a minimum value for electrons to be released.

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..... [3]

19

- (c) (i) The reading on the ammeter is proportional to the intensity of the radiation. Use your knowledge of the photoelectric effect to explain why.

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..... [3]

- (ii) When the light meter is irradiated with monochromatic radiation of frequency  $8.2 \times 10^{15}$  Hz, the number of electrons emitted every second is  $3.1 \times 10^{18} \text{ s}^{-1}$ .

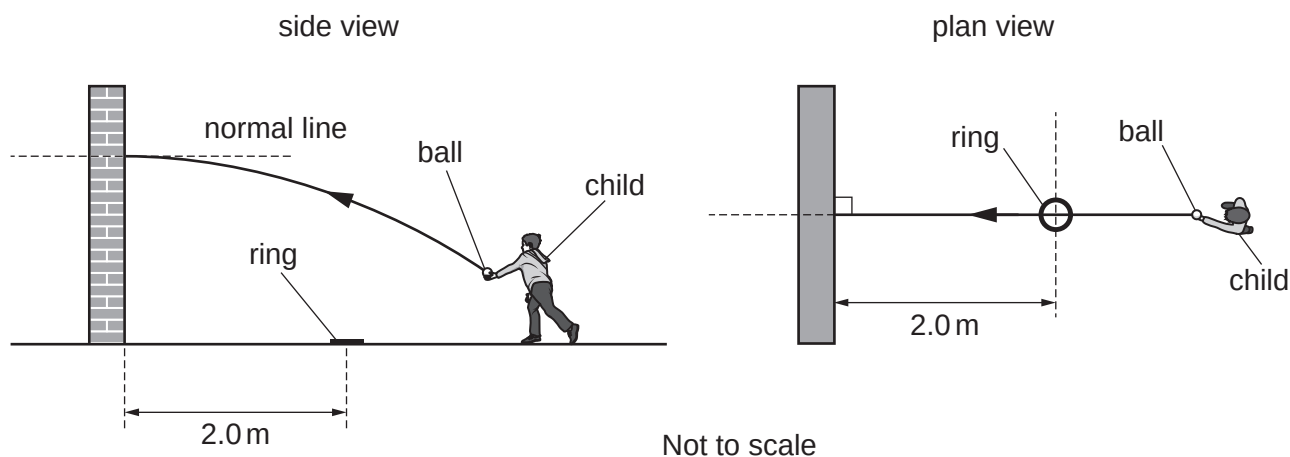
The surface area of the metal plate normal to the incident radiation is  $4.9 \times 10^{-3} \text{ m}^2$ .

Determine the intensity of the radiation.

intensity = .....  $\text{W m}^{-2}$  [4]

20

- 24 In a game, a child throws a ball at a flat, vertical wall. The ball rebounds from the wall. The child wins the game if the ball lands within a circular ring placed on the ground.



The ring has a radius of 15 cm. The centre of the ring is 2.0 m from the wall. The child throws a ball with a mass of 0.058 kg towards the wall.

- (a) The ball is incident normally on the wall with a horizontal velocity of  $7.2 \text{ m s}^{-1}$ . The ball is in contact with the wall for 52 ms before rebounding normally with a horizontal velocity of  $3.6 \text{ m s}^{-1}$ .

Calculate the average magnitude of the force that the wall exerts on the ball.

force = ..... N [3]

- (b) The ball lands on the ground a time of 0.58 s after hitting the wall.

Show that the child wins the game. Air resistance can be assumed to be negligible.

[3]

## 21

- 25** A soil scientist investigates how different types of soil particles fall through water.

A soil scientist measures the terminal velocity of soil particles in water. He fills a tall glass cylinder with water and places a small sample of soil into the water. He uses a video camera, with a known frame rate, to measure the time taken for a particle in the soil to fall a measured distance.

The scientist records the total distance that the particle falls every 0.1 s.

| Time/s | Total distance/cm |
|--------|-------------------|
| 0.0    | 0.0               |
| 0.1    | 1.2               |
| 0.2    | 3.5               |
| 0.3    | 6.2               |
| 0.4    | 8.9               |
| 0.5    | 11.6              |

- (a)** Use the information in the table to explain why using a video camera to measure the time is more appropriate than a stopwatch in this investigation.

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..... [2]

- (b)** Suggest one other precaution that the scientist should take to ensure that the terminal velocity is determined as accurately as possible.

.....

..... [1]

- (c)** Use the scientist's results to show that the terminal velocity is about  $0.3 \text{ m s}^{-1}$ .

[2]

- (d) Soil scientists classify soil samples according to the diameter of the particles, as shown in the table below.

| Soil sample | Range of diameter /mm |
|-------------|-----------------------|
| clay        | < 0.002               |
| silt        | 0.002 – 0.05          |
| sand        | 0.05 – 2.0            |
| gravel      | >2.0                  |

The terminal velocity of a spherical particle of radius  $r$ , falling in water is given by

$$v = \frac{2r^2 g(\rho_s - \rho_w)}{9\eta}$$

$\rho_w$  = density of water =  $1000 \text{ kg m}^{-3}$

$\rho_s$  = density of soil particle =  $1500 \text{ kg m}^{-3}$

$\eta$  = constant for water =  $1.0 \times 10^{-3} \text{ Pa s}$

$g$  = acceleration of free fall =  $9.81 \text{ ms}^{-2}$

Determine which soil sample was used by the scientist in this investigation.

soil sample = ..... [3]

**END OF QUESTION PAPER**



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